

2025 Science Olympiad CSE - Chemistry Lab

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Name: KEY

School/Team Number: KEY

Score: 55 /55

Instructions:

- ➢ Write your full name, school, and team number in the spaces provided above.
- ➢ Only a periodic table and a non-programmable scientific calculator are allowed.
- ➢ Assume ideal conditions unless otherwise stated.
- ➢ Round answers to the correct number of sigfigs and include appropriate units.
- ➢ Partial credit will be awarded based on work shown.

1) [1 point] A 0.450 g sample of X_2CO_3 is titrated with 42.86 mL of 0.250 M HCl. If the carbonate is the only basic component, what is the molar mass of the metal cation M?

   a) 10.5 g/mol b) 11.0 g/mol c) 12.0 g/mol d) 12.5 g/mol

2) [1 point] Which of the following units is the rate constant k of a fifth-order reaction?

   a) $mol^{-3}L^3s^{-1}$ b) $mol^{-4}L^4s^{-1}$ c) $mol^{-5}L^5s^{-1}$ d) $mol^{-6}L^6s^{-1}$

3) [1 point] A sample of impure potassium chlorate $(KClO_3)$ decomposes as follows:

$$2\ KClO_3 \rightarrow 2\ KCl + 3O_2$$

   If heating 3.00 g of the sample produces 0.672 L of O_2 gas at STP, what is the mass percent of pure $KClO_3$ in the sample?

   a) 80.8% b) 81.1% c) 81.4% d) 81.7%

4) [1 point] The rate determining step of a reaction is:

   a) Step with lowest activation energy b) Step with highest activation energy
   c) Fastest step d) Slowest step

5) [1 point] Hydrazine (N_2H_4) is used as a rocket propellant. It reacts to oxygen according to:

$$N_2H_4 + O_2 \rightarrow N_2 + 2\ H_2O$$

   If 10.0 g of hydrazine reacts completely, what volume of water vapor is produced at 275 °C and 1.27 atm?

   a) 11 L b) 12 L c) 22 L d) 23 L

6) [1 point] A reaction follows the rate law $Rate = k[A][B]^2$. If the concentration of A is doubled and the concentration of B is halved, by what factor will the rate be multiplied by?

   a) 1/2 b) 1 c) 2 d) 4

7) [1 point] In acidic solution, potassium permanganate ($KMnO_4$) reacts with sodium sulfite (Na_2SO_3) to form manganese (II) sulfate ($MnSO_4$), sodium sulfate (Na_2SO_4), and water. What is the sum of the coefficients for the balanced net ionic equation?

a) 23 b) 25 c) 27 d) 29

8) [1 point] Which of the following quantities affect the rate constant of a reaction?

I) Concentration of the reactants

II) Temperature

III) Activation Energy

a) I and II b) I and III c) II and III d) I, II, and III

9) [1 point] Acetylene is prepared by dropping water onto solid calcium carbide:

$$CaC_2(s) + 2H_2O(l) \rightarrow C_2H_2(g) + Ca(OH)_2(s)$$

A cylinder needs to be filled with 455 g of acetylene. What minimum mass of CaC_2 (kg) is needed if the process operates at 85% of the theoretical yield?

a) 1.13 kg b) 1.18 kg c) 1.25 kg d) 1.32 kg

10) [1 point] What is the difference between an intermediate and transition state?

a) An intermediate is a species that exists momentarily and is formed during the reaction, while a transition state is a high-energy, unstable configuration that exists only at the peak of the activation energy barrier.

b) An intermediate is a high-energy, unstable species, while a transition state is a stable, isolable species.

c) An intermediate is formed after the products are generated, while a transition state occurs only after the reactants are consumed.

d) An intermediate is the final product of the reaction, while a transition state is the species that initiates the reaction.

11) [1 point] A 4.18 g sample of a compound contains 0.80 g of carbon, 0.20 g of hydrogen, and the rest oxygen. What is the empirical formula?

a) CH_3O_3 b) $C_2H_4O_3$ c) $C_3H_8O_4$ d) $NOTA$

12) [1 point] What is the effect of increasing the concentration of a reactant on the rate of reaction for a first-order reaction?

a) Rate increases b) Rate decreases
c) Rate stays the same d) Cannot be determined with information given

13) [1 point] What principle in kinetics states that the transition state of a reaction resembles the structure of the reactants or products that are closest in energy?

a) Arrhenius Equation b) Hammond's Postulate
c) Collision Theory d) Transition State Theory

14) [1 point] A plot of $ln[A]$ vs time yields a straight line with slope -0.00693. What is the half-life of this reaction?

a) 0.01 s b) 10 s c) 100 s d) 1000 s

15) [1 point] An unknown hydrate $MgSO_4 \cdot xH_2O$ is heated. The initial mass is 6.123 g, and after heating to remove water, the residue weighs 2.984 g. What is the value of x?

a) 4 b) 5 c) 6 d) 7

16) [1 point] The following mechanism is proposed for a reaction:

Step 1: $NO_2 + F_2 \leftrightarrow NO_2F + F$ (fast equilibrium)

Step 2: $F + NO_2 \rightarrow NO_2F$ (slow)

What is the rate law for the overall reaction?

a) $rate = k[NO_2][F_2]$ b) $rate = k[NO_2]^2[F_2]$
c) $rate = k[NO_2]^2$ d) $rate = k[NO_2][NO_2F\}$

17) [1 point] Which of the following are oxidation-reduction reactions?

I) $2Mg\ (s)\ +\ O_2\ (g) \rightarrow 2MgO\ (s)$

II) $HCl\ (aq)\ +\ NaOH\ (aq) \rightarrow NaCl\ (aq)\ +\ H_2O\ (l)$

III) $Zn\ (s)\ +\ Cu^{2+}\ (aq) \rightarrow Zn^{2+}\ (aq)\ +\ Cu\ (s)$

IV) $CaCO_3\ (s) \rightarrow CaO\ (s)\ +\ CO_2\ (g)$

a) I & II b) I & III c) II & IV d) III & IV

18) [1 point] A 25.0 g sample of potassium chlorate $(KClO_3)$ decomposes completely in heat:

$$2KClO_3\ (s) \rightarrow 2KCl\ (s)\ +\ 3O_2\ (g)$$

If the decomposition is 80.0% efficient, how many liters of oxygen gas (at STP) is collected?

a) 3.73 L b) 4.48 L c) 6.00 L d) 8.21 L

19) [1 point] A student combusts 1.20 g of pure propane gas (C_3H_8) in excess oxygen. What is the total number of moles of gaseous products formed?

a) 0.0273 mol b) 0.0818 mol c) 0.1090 mol d) 0.136 mol

20) [1 point] Consider the following reaction:

$$2H_2O_2\ (aq) \rightarrow 2H_2O\ (l)\ +\ O_2\ (g)$$

How many of the following correctly describe this reaction?

I. Decomposition Reaction
II. Oxidation-Reduction Reaction
III. Combustion Reaction
IV. Acid-Base Reaction

a) 1 b) 2 c) 3 d) 4

21) [2 points] A yellow compound with a total mass of 5.82 grams is analyzed and found to contain 2.339 grams of potassium (K), 1.551 grams of chromium (Cr) and the rest oxygen (O). What is the empirical formula of this compound?

5.82 - 2.339 - 1.551 = 1.93 g

Potassium (K): 2.339 g, 0.0598 mol

Chromium (Cr): 1.551 g, 0.0298 mol

Oxygen (O): 1.93 g, 0.1206 mol [1]

K: 0.0598/0.0298 = 2

Cr: 0.0298/0.0298 = 1

O: 0.1206/0.0298 = 4

K2CrO4 , Potassium Chromate [1]

22) [3 points] In acidic solution, the yellow compound from Question #21 reacts with ethanol (C_2H_5OH). Ethanol is oxidized to acetic acid (CH_3COOH), and the anion from the compound is reduced to Cr^{3+}. Write the balanced net ionic equation for this reaction:

Find both half reactions:

Oxidation: $C_2H_5OH + H_2O \rightarrow CH_3COOH + 4H^+ + 4e^-$ [1]

Reduction: $CrO_4^{2-} + 8H^+ + 3e^- \rightarrow Cr^{3+} + 4H_2O$ [1]

Multiply oxidation half reaction by 3, multiply reduction half reaction by 4 and simplify

$4CrO_4^{2-} + 3C_2H_5OH + 20H^+ \rightarrow 4Cr^{3+} + 3CH_3COOH + 13H_2O$ [1]

23) [3 points] If 2.00 g of (Question #18) and 1.00 g of ethanol (C_2H_5OH) are mixed in excess acid, determine the limiting reagent [1]. Calculate the maximum mass [1] of acetic acid (CH_3COOH) that can be produced from the limiting reagent, and find how many grams of the excess reagent [1] remain after the reaction is complete.

Moles K_2CrO_4 = 2.00/194.20 = 0.0103 mol, Moles C_2H_5OH = 1.00/46.07 = 0.0217

Mole ratio = 4/3, so chromate is limiting reagent. [1] 4 moles of chromate = 3 moles acetic acid, 0.0103 x 3/4 = 0.0077 mol acetic acid. [1] Ethanol moles used = 0.0103 x ¾ = 0.0077 mol, 0.0217 - 0.0077 = 0.0140 mol = 0.644 g. [1]

24) The decomposition of dinitrogen pentoxide in a solution occurs as follows:

$$2N_2O_5 \rightarrow 4NO_2 + O_2$$

A student conducts the decomposition reaction at $45\ C°$ and measures the concentration of N_2O_5 at various times:

$Time\ (s)$	$[N_2O_5]$
0	0.800
100	0.567
200	0.402
300	0.285
400	0.202
500	0.143

a) [2 points] Using the data provided, determine the order of the reaction with respect to N2O5. Show all work and calculation.

Solution should include a table with time and $ln[N_2O_5]$ values [1]. It should also show between 2 time intervals (0-100, 100-200, etc) the difference in $ln[N_2O_5]$ is the same (around -0.3459). This indicates that the reaction is first order [1] with respect to N_2O_5.

b) [3 points] Calculate rate constant k for this reaction and include appropriate units.

The first-order integrated rate law is $ln[N_2O_5] = ln[N_2O_5]_0 - kt$ [2]. You can solve for k by plugging in any data point, resulting in $k = 0.00345[1]\ s^{-1}[1]$.

c) [2 points] Use the integrated rate law for the reaction and predict the concentration of N_2O_5 at $t = 750\ s$.

Using the first-order integrated rate law again, $ln[N_2O_5] = ln[N_2O_5]_0 - kt$ [1]

$ln[N_2O_5] = ln(0.800) - (0.00345)(750).\ [N_2O_5] = e^{-2.803} = 0.0606\ M$ [1].

2nd point can still be earned even if you get a wrong value for part b), as long as the remaining calculation is right.

25) Another student proposes this two-step mechanism for the decomposition of N_2O_5:

$$Step\ 1\ (fast\ equilibrium):\ N_2O_5 \leftrightarrow NO_2 + NO_3$$

$$Step\ 2\ (slow):\ NO_2 + NO_3 \rightarrow NO_2 + O_2 + NO$$

a) [2 points] Write the overall reaction for this mechanism.

Cancel the intermediates $(NO_3\ and\ NO_2)$ that appear on both sides [1], which results in an overall reaction of $N_2O_5 \rightarrow NO_2 + O_2 + NO$ [1]

b) [2 points] Is this mechanism consistent with the observed rate law? (points will only be given if a justification is provided).

From earlier, we determined this reaction is first order. Step 2 is slow, so it determines the rate. Rate law based on Step 2 is $Rate = k[NO_2][NO_3]$. However, both are intermediates, not reactants. Since Step 1 is fast equilibrium, we can use it to express the concentration of the intermediates in terms of the stable species.

$N_2O_5 \leftrightarrow NO_2 + NO_3 \rightarrow K = [NO_2][NO_3]/[N_2O_5] \rightarrow [NO_2][NO_3] = K[N_2O_5]$. [1]

Substituting back, $Rate = k[NO_2][NO_3] = kK[N_2O_5]$. Since kK is a constant, the rate law simplifies to $Rate = k'[N_2O_5]$, which is consistent with the observed rate law. [1]

c) [2 points] If your answer to the previous question is no, propose a modification or alternate/extra step that would better align with the observed rate law. Put N/A if the mechanism is already consistent with the observed rate law.

N/A

d) [3 points] A third student repeats the experiment at a double the original temperature in Kelvin and observes that the rate constant increases by a factor of 20000. Estimate the activation energy for the decomposition reaction.

Arrhenius Equation: $ln(\frac{k_2}{k_1}) = \frac{E_a}{R}(\frac{1}{T_1} - \frac{1}{T_2})$ [1] We are given $k_2/k_1 = 20000$, $T_1 = 318\ K$, $T_2 = 636\ K$, and $R = 8.314\ J/mol \cdot K$.

$ln(20000) = \frac{E_a}{8.314}(\frac{1}{318} - \frac{1}{636})$ [1], $E_a = 52.4\ kJ/mol.$ [1]

26) I is an intermediate for the reaction below, and the rate law is $Rate = k[A]^2[B]$.

$$Step\ 1:\ A + B \Leftrightarrow I \text{ (fast, rate constants } k_1 \text{ and } k_{-1})$$

$$Step\ 2:\ I + A \rightarrow P + D \text{ (slow, rate constant given by } k_2)$$

a) [2 points] Which step is the rate determining step and why?

Step 2 [1], because it is the slow and thus rate determining step. [1]

b) [4 points] Write the steady state approximation for the intermediate I, and derive an expression for $[I]$ in terms of $[A]$, $[B]$, and the rate constants.

$\frac{d[I]}{dt} = k_1[A][B] - k_{-1}[I] - k_2[I][A] \approx 0$ [2], isolate I:

$k_1[A][B] = [I](k_{-1} + k_2[A]),\ I = \frac{k_1[A][B]}{k_{-1} + k_2[A]}$ [2]

c) [3 points] Using your derived expression for $[I]$, derive the overall rate law for the formation of the product P, and show that it matches the experimentally determined rate law. Assume that k_2 is incredibly small compared to k_1 or k_{-1}.

$Rate:\ v = \frac{d[P]}{dt} = k_2[I][A]$, [1] substitute the steady state approximation from part b),

$v = k_2[A] \cdot \frac{k_1[A][B]}{k_{-1} + k_2[A]} = \frac{k_1 k_2[A]^2[B]}{k_{-1} + k_2[A]}$ [1]

$k_{-1} >> k_2[A],\ thus\ v = \frac{k_1 k_2}{k_{-1}}[A]^2[B]$ [1]

This matches the experimental form, $Rate = k[A]^2[B]$ with $k \approx \frac{k_1 k_2}{k_{-1}}$

d) [2 points] If the equilibrium constant for step 1 of the reaction is given by $K = k_1/k_{-1}$, rewrite the overall rate constant k in terms of K and k_2.

$Rate = k[A]^2[B]$, using part c), $k = \frac{k_1 k_2}{k_{-1}}$ [1]

$K = \frac{k_1}{k_{-1}}$

$k = Kk_2$ [1]